

QBUS2820 Predictive Analytics, S2 2026

Week 6&7: Advanced regression (Shrinkage methods)

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Recommended reading

- Section 6.2, *An Introduction to Statistical Learning with Applications in R* by James et al.: easy to read, sloppy discussions sometimes, comes with R/Python code for practice.
- Section 3.4, *The Elements of Statistical Learning* by Hastie et al.: well-written, deep in theory, suitable for students with a sound maths background.

Ridge Regression

Multiple linear regression (recall)

Recall the linear model

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \epsilon_i = x_i' \beta + \epsilon_i \quad i = 1, \dots, n,$$

where $\mathbb{E}(\epsilon_i) = 0$, $\mathbb{V}(\epsilon_i) = \sigma^2$. Let $\beta = (\beta_0, \dots, \beta_p)' \in \mathbb{R}^{p+1}$.

$$X = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{i1} & \cdots & x_{ip} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_i \\ \vdots \\ y_n \end{bmatrix}, \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_i \\ \vdots \\ \epsilon_n \end{bmatrix}.$$

Matrix form:

$$\mathbf{y} = X\beta + \boldsymbol{\epsilon}$$

The ordinary least squares (OLS) method estimate of β

$$\hat{\beta} = (X'X)^{-1}X'y,$$

given that the inverse of matrix $X'X$ exists.

What if the inverse of matrix $X'X$ doesn't exist, or near singular?

Norms

Let $\mathbf{a} = (a_1, \dots, a_p)'$.

- ℓ_0 **norm**: the number of non-zero elements in the vector.

$$\|\mathbf{a}\|_0 = \sum_{i=1}^p \mathbf{1}(a_i \neq 0)$$

- ℓ_1 **norm**: the sum of the absolute values of its elements.

$$\|\mathbf{a}\|_1 = \sum_{i=1}^p |a_i|$$

- ℓ_2 **norm**: the square root of the sum of the squared elements of $\mathbf{a}$.

$$\|\mathbf{a}\| = \sqrt{\mathbf{a}'\mathbf{a}} = \sqrt{\sum_{i=1}^p a_i^2}$$

Positive semidefinite matrix*

A symmetric matrix $\mathbf{A} \in \mathbb{R}^{p \times p}$ is positive semidefinite if for all non-zero vectors $\mathbf{b} \in \mathbb{R}^p$, the following condition holds:

$$\mathbf{b}^\top \mathbf{A} \mathbf{b} \geq 0$$

Eigendecomposition*

- For a symmetric positive semidefinite matrix $\mathbf{A} \in \mathbb{R}^{p \times p}$, the eigendecomposition is given by:

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}'$$

- $\mathbf{Q}$ is an $p \times p$ orthogonal matrix whose columns are the eigenvectors of $\mathbf{A}$ with $\mathbf{Q}'\mathbf{Q} = \mathbf{Q}\mathbf{Q}' = \mathbf{I}_p$
 - $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_p)$ is an $p \times p$ diagonal matrix whose diagonal elements are the eigenvalues of $\mathbf{A}$.
- Key Properties:**
 - All eigenvalues in $\mathbf{\Lambda}$ are non-negative, i.e., $\lambda_i \geq 0$ for all i .
 - $\mathbf{A}^{-1} = \mathbf{Q}(\mathbf{\Lambda})^{-1}\mathbf{Q}^{-1} = \mathbf{Q} \cdot \text{diag}(1/\lambda_1, \dots, 1/\lambda_p) \cdot \mathbf{Q}^{-1}$.
 - $\text{tr}(\mathbf{A}) = \sum_{i=1}^p \lambda_i$, where $\text{tr}(A)$ denotes the **trace** of matrix A , which is the sum of the diagonal elements of A .

Multicollinearity

Multicollinearity issue: when at least two columns of X are strongly correlated to each other, then $X'X$ is nearly singular.

- $\text{tr}((X'X)^{-1})$ will be large.
- It can be shown that the expected squared error
$$\mathbb{E}[\|\hat{\beta} - \beta\|^2] = \mathbb{E}[(\hat{\beta} - \beta)'(\hat{\beta} - \beta)] = \sigma^2 \text{tr}((X'X)^{-1}).$$
- Estimation is unstable: a small change in data might lead to a large change in the estimate $\hat{\beta}$, even in signs, and some components of $\hat{\beta}$ might be extremely large in absolute value

Multicollinearity*

- $X'X$ is a symmetric positive semidefinite matrix.
- Let $d_1^2, \dots, d_p^2$ be the eigenvalues of $X'X$. Then $1/d_1^2, \dots, 1/d_p^2$ be the eigenvalues of $(X'X)^{-1}$.
- If $X'X$ is near singular, then some of the d_i^2 will be close to zero.
- The trace of a matrix is the sum of its eigenvalues, hence

$$\text{tr}((X'X)^{-1}) = \sum_{i=1}^p \frac{1}{d_i^2}.$$

- Hence, for multicollinearity situations,

$$\mathbb{E}\|\hat{\beta} - \beta\|^2 = \sigma^2 \sum_{i=1}^p \frac{1}{d_i^2}$$

can be large.

Multicollinearity

Typical warning from statistical software: **Warning: Matrix is close to singular or badly scaled. Results may be inaccurate. RCOND = 1.012021e-16.**

- Consider model 1:

$$\text{AmountSpent} = \beta_0 + \beta_1 \times \text{Salary} + \beta_2 \times \text{Calalogs} + \epsilon.$$

$$\text{tr}((X'X)^{-1}) = 0.008$$

- Model 2:

$$\text{AmountSpent} = \beta_0 + \beta_1 \times \text{Salary} + \beta_2 \times \text{Income} + \epsilon.$$

$$\text{tr}((X'X)^{-1}) = 992.77$$

Ridge regression

- Shrinkage methods fit the model containing all p predictors using the approaches that regularizes the coefficient estimates, or i.e. shrinks the estimated coefficients towards zero.
- Ridge regression is motivated from removing the multicollinearity effect. Use the white board.
- Shrinkage method are developed to reduce the large variance.

Ridge regression

- The least square estimate $\beta_0, \beta_1, \dots, \beta_p$ through minimizing

$$RSS = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2.$$

- Ridge regression estimates $\hat{\beta}_\lambda^R$ through minimizing

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 = RSS + Penalty,$$

where $\lambda \geq 0$ is the regularization parameter, to be determined separately.

- Larger or smaller λ ? Selecting a good λ is very important, cross validation or AIC/BIC can be used for this (to be discussed).
- Note the shrinkage penalty is applied to the slope **not the intercept**.

Ridge regression (Matrix representation)

Suppose each column of X and $\mathbf{y}$ have been centered to have mean zero. Then we can write ridge regression in the following matrix form

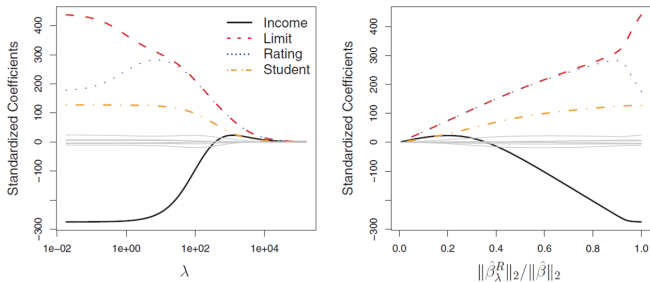
$$\hat{\beta}_{\lambda}^R = \arg \min_{\beta} \left\{ \|\mathbf{y} - X\beta\|^2 + \lambda \|\beta\|^2 \right\}$$

with X the design matrix (without the column of 1s).

- It's easy to see that $\hat{\beta}_{\lambda}^R = (X'X + \lambda I_p)^{-1} X' \mathbf{y}$.
- Theorem: There exists a $\lambda > 0$ such that

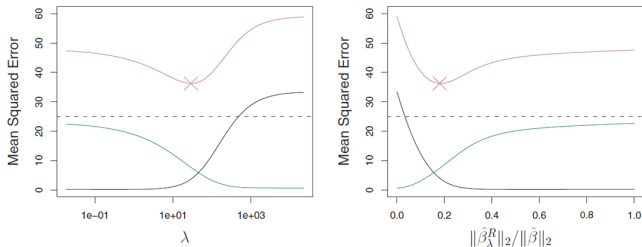
$$\mathbb{E} \|\hat{\beta}_{\lambda}^R - \beta\|^2 < \mathbb{E} \|\hat{\beta} - \beta\|^2$$

Credit data example (solution paths)



- Left: each curve corresponds to the ridge regression coefficient estimate for one of ten variables.
- Right: instead of plotting vs the regularization parameter, we display $\|\hat{\beta}_\lambda^R\|_2 / \|\hat{\beta}\|_2$, where $\hat{\beta}$ denotes the OLS.

Bias and variance tradeoff for ridge regression



Simulated data with $n = 50$ observations and $p = 45$ predictors all having nonzero coefficients. Squared bias (black), variance (green) and test MSE (purple) for the ridge regression predictions on a simulated data set. The horizontal dashed lines indicate the minimum possible MSE.

Scaling of predictors in ridge regression

- The standard OLS are **scale-invariant**, multiplying $\mathbf{X}_i$'s by a constant c will lead to the OLS by a factor of $1/c$. The prediction for the test data $\tilde{\mathbf{X}}_i' \hat{\beta}$ remains the same.
- Ridge regression coefficient estimates can change significantly when a predictor is multiplied by a constant. (**Scale-invariant or not? Why?**)
- Hence it is best to perform **standardization for the predictors** before applying ridge regression,

$$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}} \quad \text{or} \quad \tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{\sqrt{\frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}}.$$

Ridge regression: the selection of λ

Selecting λ ?

- Let $d_1^2, \dots, d_p^2$ be the eigenvalues of $X'X$.
- It's argued that the **degrees of freedom** defined as

$$\text{df}(\lambda) = \sum_{j=1}^p \frac{d_j^2}{d_j^2 + \lambda}$$

can be used to characterise the number of parameters.

- As $\lambda = 0$ (no regularisation, this is OLS regression),
 $\text{df}(\lambda) = p$.
- As $\lambda \rightarrow \infty$, $\text{df}(\lambda) \rightarrow 0$.

Ridge regression: the selection of λ

Let $\hat{\sigma}_\lambda^2 = \|\mathbf{y} - X\hat{\beta}_\lambda^R\|^2/n$.

$$\text{AIC}(\lambda) = n \log(\hat{\sigma}_\lambda^2) + 2 \times \text{df}(\lambda)$$

$$\text{BIC}(\lambda) = n \log(\hat{\sigma}_\lambda^2) + \log(n) \times \text{df}(\lambda)$$

We can chose λ that minimises $\text{AIC}(\lambda)$, $\text{BIC}(\lambda)$ or $\text{CV}(\lambda)$.

Example: Baton Rouge house price

- Consider predicting the house price. Let's use the first 900 observations as training, the rest for testing.
- To be simple, variable "Style" is removed as it has 12 categories (would need 11 dummy variables to represent, too much!)
- The training output data is centered to have mean 0, and the predictors are standardized
- Note: the test data should be centered/standardized respectively

Example: Baton Rouge house price

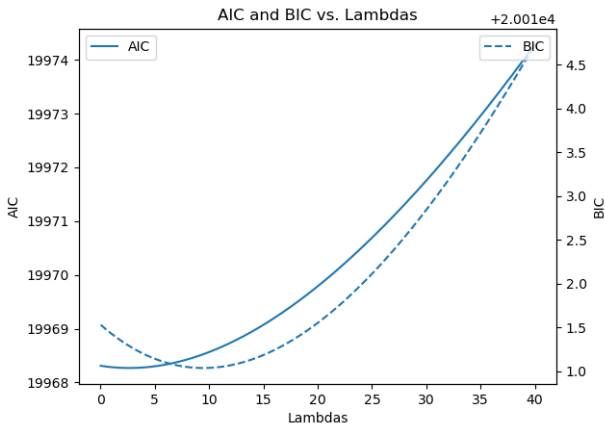


Figure 1: The plot of AIC and BIC as λ varies for the Baton Rouge house price example

Example: Baton Rouge house price

	λ chosen	Test error
OLS	N/A	1.1137e+05
AIC	2.64	1.1133e+05
BIC	9.47	1.1127e+05

The Lasso and its Variants

The lasso

- Ridge regression still include all p predictors in the model. (No variable selection!)
- The lasso ([Robert Tibshirani, 1996, Journal of Royal Statistical Society: Series B](#)) was proposed to overcome this advantage.
- The lasso coefficient $\hat{\beta}_{\lambda}^L$

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| = RSS + Penalty,$$

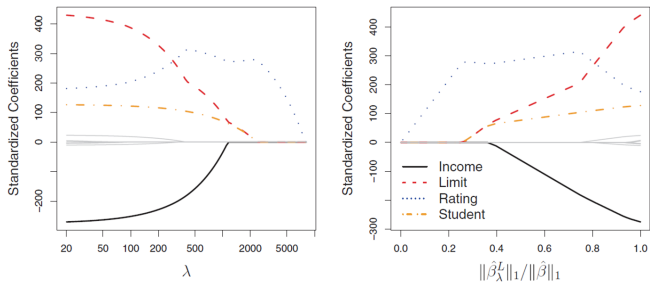
where $\lambda \geq 0$ is the regularization parameter.

- Ridge regression considers ℓ_2 type of norm for the penalty term and the lasso uses a ℓ_1 norm of penalty.

The lasso

- The lasso also shrinks the coefficient estimates towards zero.
- However, ℓ_1 penalty has the effect of forcing some of the coefficients to be exactly zero where a large λ is used.
- The lasso can perform **variable selection** and choose a **sparse** model containing a subset of all p predictors.
- What about the selection of regularization parameter? CV or AIC/BIC.

Credit data example (solution paths)



Selecting λ

The shrinkage parameter λ can be selected using the BIC/AIC criterion or cross-validation method.

Let $\mathbf{X}$ be the design matrix and $\mathbf{y}$ be the vector of responses. Denote by $\hat{\beta}_{\lambda}^L$ the Lasso estimate of β given λ . Define

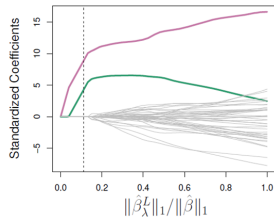
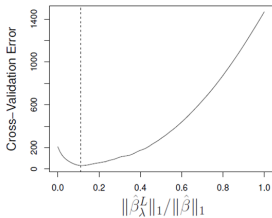
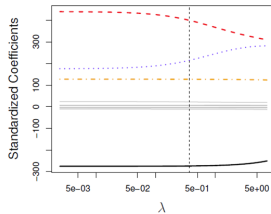
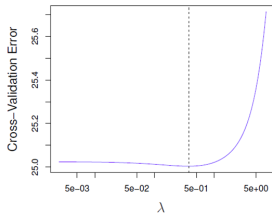
$$\text{BIC}(\lambda) = n \log \left(\frac{\|\mathbf{y} - \mathbf{X} \hat{\beta}_{\lambda}^L\|^2}{n} \right) + \log(n) \times df_{\lambda},$$

$$\text{AIC}(\lambda) = n \log \left(\frac{\|\mathbf{y} - \mathbf{X} \hat{\beta}_{\lambda}^L\|^2}{n} \right) + 2 \times df_{\lambda}.$$

df_{λ} is called the **degrees of freedom**, which is approximately the number of non-zero coefficients in the model.

The selected λ is the one that minimizes $\text{BIC}(\lambda)$, $\text{AIC}(\lambda)$ or the cross-validation error, defined as before.

Credit data example and simulation 2



Comparing the Lasso and Ridge Regression

Parameter selection for ridge and lasso regression

- **Bias and variance tradeoff:** As λ increase, the model becomes sparser, so that the variance will decrease and the bias will increase. The test MSE will behave like a U shape.
- We choose a grid of λ values and compute the AIC, BIC or CV errors for each λ .
- We then select the regularization parameter λ for which the preferred error criteria is the smallest.
- Finally, the model is **re-fitted** using all of the available observations.
 - **Ridge:** refit with the **selected values of the regularization parameter**;
 - **Lasso:** refit an ordinary least squares (OLS) regression model using only these **selected predictors**. (optional)

Another formulation for ridge regression and the lasso

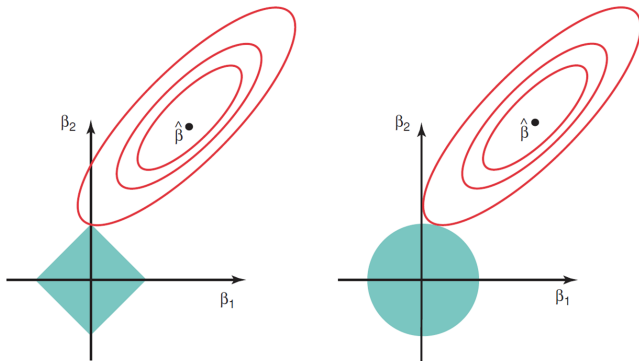
- One can easily show that the lasso and ridge regression coefficient estimates solve the following problems

$$\min_{\beta} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq s$$

and

$$\min_{\beta} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq s.$$

A graphical illustration



The solid blue areas are the constraint regions, $|\beta_1| + |\beta_2| \leq s$ and $\beta_1^2 + \beta_2^2 \leq s$, while the red ellipses are the contours of the RSS.

A simple special case for ridge regression and the lasso

- Consider a simple special case $n = p$ and the design matrix $\mathbf{X}$ is an identity matrix. Also consider the regression without an intercept.
- Least squares** aim to minimize $\sum_{j=1}^p (y_j - \beta_j)^2$ and the solution is given by

$$\hat{\beta}_j = y_j, j = 1, \dots, p.$$

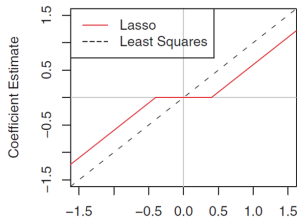
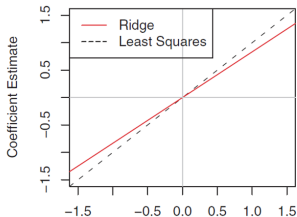
- Ridge regression** considers minimizing $\sum_{j=1}^p (y_j - \beta_j)^2 + \lambda \sum_{j=1}^p \beta_j^2$ and the solution is given by

$$\hat{\beta}_j^R = y_j / (1 + \lambda)$$

- The lasso** considers minimizing $\sum_{j=1}^p (y_j - \beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j|$ and the solution is given by

$$\hat{\beta}_j^L = \begin{cases} y_j - \lambda/2 & \text{if } y_j > \lambda/2 \\ y_j + \lambda/2 & \text{if } y_j < -\lambda/2 \\ 0 & \text{if } |y_j| \leq \lambda/2 \end{cases}$$

Graph illustration



- Ridge regression more or less shrinkage every dimension of the data by the same proportion.
- The lasso more or less shrinks all coefficients towards zero by a similar amount and sufficiently small coefficients are shrunk all the way to zero. (**soft thresholding**). **Any hard thresholding?**

Which method to use?

- Recall the **no free lunch theorem**: neither ridge regression or the lasso universally outperform the other. Both methods can improve predictive accuracy compared to OLS by **reducing overfitting**, but the specific gains depend on the data structure.
- **Lasso**:
 - **sparse solution** (particularly useful for feature selection when $p \gg n$);
 - better interpretability.
- **Ridge**:
 - handle **multicollinearity** well by distributing the coefficient values among the correlated predictors;
 - suitable if you believe most predictors have some effect, even if small.

Comparison with best subset selection

- Like the constraint versions of ridge regression and the lasso, the best subset selection can be thought of the following constraint problem

$$\min_{\beta} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^p I(\beta_j \neq 0) \leq s.$$

- ℓ_0 type of constraint.
- Unfortunately, solving this problem is computationally infeasible when p is large, since it requires considering all models containing s predictors.

ℓ_0 , ℓ_1 and ℓ_2 penalty terms

- The lasso is the only one that provides a **sparse** solution (variable selection) and also a **convex** optimization problem.
- **The lasso:** ℓ_1 penalty, shrinks all coefficients toward zero by a similar amount (**Biased estimators for large coefficients!**).
- **Best subset selection:** ℓ_0 penalty (hard thresholding), but **computationally infeasible** when p is large.
- **Ridge regression:** ℓ_2 penalty (**no sparsity!**).
- **Combine some penalties?**

Group lasso penalty

- Suppose p predictors are divided into G groups, with p_g the size in group $g, g = 1, \dots, G$. The group lasso minimizes the following convex criterion

$$\sum_{i=1}^n \left(y_i - \beta_0 - \sum_{g=1}^G \mathbf{x}_{ig}^T \boldsymbol{\beta}_g \right)^2 + \lambda \sum_{g=1}^G \|\boldsymbol{\beta}_g\|,$$

where $\mathbf{x}_{ig} \in \mathbb{R}^{p_g}, i = 1, \dots, n$ denote the predictors within g -th group with the corresponding coefficient vector $\boldsymbol{\beta}_g \in \mathbb{R}^{p_g}$.

- Since ℓ_2 norm of a vector is zero if and only if all its components are zero, this procedure encourages sparsity at both group and individual levels.
- Group lasso penalty enforces **group sparsity** (all within-group components are zero.)

Group lasso penalty

- For example, a categorical variable X with K categories is represented by $K - 1$ dummy variables. If we want to remove X , we need to remove all of the associated $K - 1$ coefficients together.
- Suppose that $(\beta_1, \dots, \beta_p)$ is grouped into m groups $\beta_1, \beta_2, \dots, \beta_m$. **Group Lasso** minimizes

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_p x_{ip})^2 + \lambda \sum_{j=1}^m \|\beta_j\|$$

where $\|\beta_j\|$ is the l_2 -norm of vector β_j .

Elastic Net

The **elastic net** is a compromise between ridge regression and the lasso:

$$\hat{\beta}_{\text{EN}} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p \left(\alpha \beta_j^2 + (1 - \alpha) |\beta_j| \right),$$

for $\lambda \geq 0$ and $0 < \alpha < 1$.

The elastic net performs variable selection like the Lasso, and shrinks together the coefficients of correlated predictors like Ridge regression.

Review questions

- What is the multicollinearity issue, and when might we encounter that issue?
- What are the Ridge and Lasso regressions?
- How to choose the tuning parameters in Ridge and Lasso?
- What are the penalty terms in the Ridge and Lasso methods?
- What feature makes Lasso a variable selection method?
- Why do we need to **standardize the data** before using Ridge or Lasso? How do we deal with **dummy variables** in the context of Lasso regression?